



NSAT MOCK – 1

Sr. No.	Section Name		No. of Questions
1	General Aptitude	Logic & Data Interpretation	20
		Algorithmic Thinking	10
2	English	Reading Comprehension	10
		Language Reasoning	10
3	Mathematics	10 th Class Maths	20
		11 th & 12 th Class Maths	10



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SECTION – 1: GENERAL APTITUDE

Directions:

Read the following information carefully and answer the question given below. Six students A, B, C, D, E and F took Maths test of 10 marks and each one of them got a different score. All scores were in whole numbers. None of them scored 100% marks. F scored more marks than E. B scored 4 marks more than A. D scored 7 marks. Only C scored less than 50% marks.

1. Who scored the maximum marks?
(1) B (2) C (3) D (4) E (5) F

Directions for questions 2 – 3:

Study the following information and answer the questions. In a group of six friends, 4 friends are in the math club, 2 are in the debate team, one is in the chess club and three are cheerleaders. Inder and Anjali are among the cheerleaders, while Jenny and Simran are not. Simran and Yasmin are in the debate team. Jenny, Simran, Anjali and Yasmin are all in the math club and two of them are also in the debate team. Lily is in the chess club and is not a cheerleader.

2. Who among these friends is both in the debate team and a cheerleader?
(1) Yasmin (2) Inder (3) Simran (4) Jenny
(5) None of these
3. Who among all these friends is in the math club, debate team, and is also a cheerleader?
(1) Yasmin (2) Anjali (3) Simran (4) Inder
(5) None of these

Directions:

Read the following information carefully and answer the question. Mohit is 17 m to the north of Nitin. Rohit is 15 m to the east of Mohit. Deep is 5 m to the south of Rohit. Pankaj is 4 m to the east of Deep. Rahul is 4 m to the south of Pankaj. Gagan is 6 m to the west of Rahul. Hitesh is 4 m to the north of Gagan.

4. If Vijay is 5 m to the south of Mohit, what is the distance between Pankaj and Vijay?
(1) 19 m (2) 20 m (3) 22 m (4) 23 m (5) 24 m
5. One evening, Manpreet got stuck in a traffic jam. She could see the reflection of the sun in her scooter's mirror. The traffic police diverted the traffic towards the road which was to her left. In which direction was the traffic diverted?
(1) North (2) South (3) East (4) West (5) None of these

Directions for questions 6 – 8: Study the information given below carefully and answer the questions based on it.

In a family, there are nine members – A, B, C, D, E, F, G, H and I. There are three generations and only couples have children.



A is I's grandmother. D is H's mother. F is B's son-in-law. C is E's husband. G is I's sister. D has a brother, a niece and a nephew.

6. How is H related to I's grandmother?
(1) Grandson (2) Grand-daughter (3) Son
(4) Cannot be determined (5) Daughter
7. How is F related to E's husband?
(1) Brother (2) Brother-in-law (3) Sister-in-law
(4) Son (5) Father-in-law
8. How is B related to E?
(1) Father (2) Grandfather (3) Son
(4) Father-in-law (5) Cannot be determined
9. In a certain code language, 'You Never Really Learn Much From Hearing Yourself Speak' is coded as 'From Hearing Learn Much Never Really Speak You Yourself'. How will you code 'Never Let The Fear Of Striking Out Keep You From Playing The Game'?
- (1) From Fear Game Keep Let Never Of Out Playing Striking The The You
(2) Fear From Game Keep Let Never Of Out Playing Striking The The You
(3) Fear From Game Keep Let Never Of Out Striking Playing The The You
(4) Fear From Game Let Keep Never Of Out Striking Playing The The You
(5) None of these
10. In a certain code language, if EDUCATED is coded as [2672223624] and DISCOVER is coded as [231234252040], then how will AIRPORT be coded?
- (1) [1025333435] (2) [1925333435] (3) [1925333430] (4) [1925333235]
(5) [1025323435]
11. If in a certain coded, SUGER is coded as WOZPKCJZVN, then which among the following options is coded as 'QITJQIRHCU'?
- (1) MARGIN (2) MEMES (3) MUMMY (4) MOMMY
(5) None of these

Directions for questions 12 – 14: The table given below shows the number of sold and unsold school bags manufactured by 3 small scale factories on different days of the week. Read the data carefully and answer the questions.

(Some data are missing.)

(School bags unsold on a day are put in selling stock for the next day.)

Day	A		B		C	
	Sold	Unsold	Sold	Unsold	Sold	Unsold

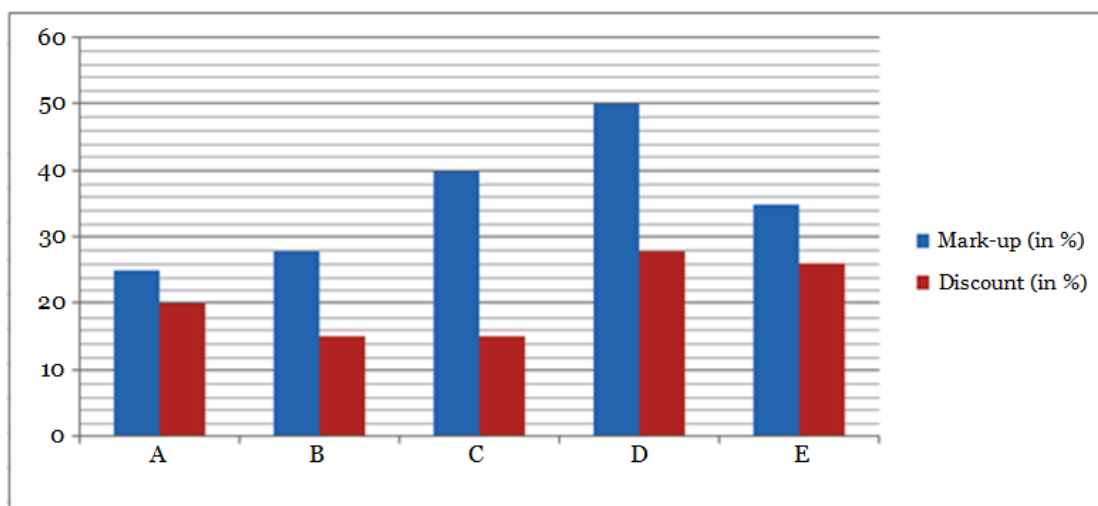


Monday	120	20	-	30	160	-
Tuesday	150	-	180	-	150	30
Wednesday	-	50	200	20	180	20

12. If the total bags manufactured by B on Tuesday were 10 less than the total bags manufactured by B on Wednesday, then what is the ratio of unsold bags on Tuesday and Wednesday, respectively?
(1) 5 : 4 (2) 3 : 2 (3) 2 : 1 (4) 4 : 3 (5) None of these
13. The total number of unsold bags of A on Tuesday and Wednesday was 20% of the total bags manufactured by A on Tuesday and Wednesday. How many bags were manufactured by A on Tuesday and Wednesday, given that the unsold bags on Tuesday were 190 fewer than the total bags sold on Wednesday?
(1) 390 (2) 400 (3) 420 (4) 450 (5) 500
14. The average of the bags manufactured by A, B and C on Wednesday was 170 and the average of the bags sold by A, B and C on Wednesday was 180. If the number of unsold bags of A on Tuesday was half the number of unsold bags of B on Tuesday, find the ratio of the unsold bags of A, B and C on Tuesday and Wednesday.
(1) 3 : 4 (2) 4 : 3 (3) 1 : 2 (4) 5 : 4 (5) 3 : 2

Directions for questions 15 – 17: Study the following information and answer the questions.

The provided bar graph displays the percentage of mark-up and discount for five different grocery items in a shop.



Selling Price = Mark-up Price – Discount (in Rs.)

Marked Price = Cost Price $\times$ (1 + mark-up%)

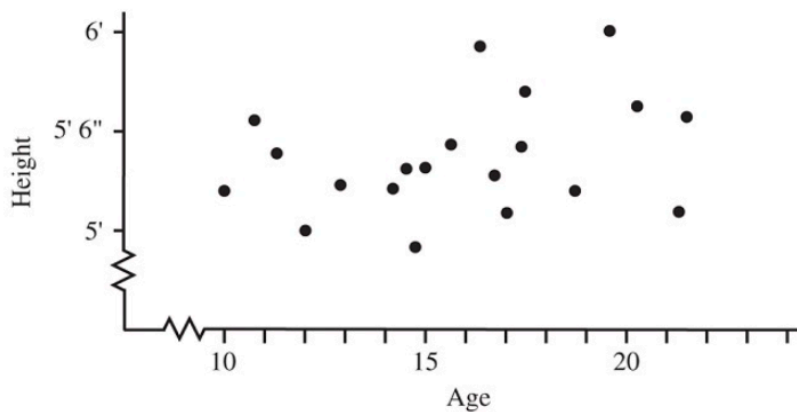
15. If the cost price of item D is Rs. 1,200, then what is the profit percentage for item D?
(1) 5 (2) 8 (3) 12 (4) 15 (5) None



16. What is the ratio of the marked price of item C to that of item D, if the selling price of item C is Rs. 510 and that of item D is Rs. 1,224?
- (1) 17 : 6 (2) 6 : 7 (3) 6 : 17 (4) 5 : 7 (5) None of these

17. If a customer buys item E at an additional discount of 13% and pays Rs. 450.66, then what is the marked price of item E?
- (1) Rs. 705 (2) Rs. 640 (3) Rs. 700 (4) Rs. 800 (5) None of these

18.



- The above scatter plot represents the ages and heights of 20 randomly selected students from class X to Class XII. What percentage of these students, in the age range of 14 to 20 years, have height under 5 feet 6 inches?
- (1) 37% (2) 41% (3) 45% (4) 47% (5) 53%

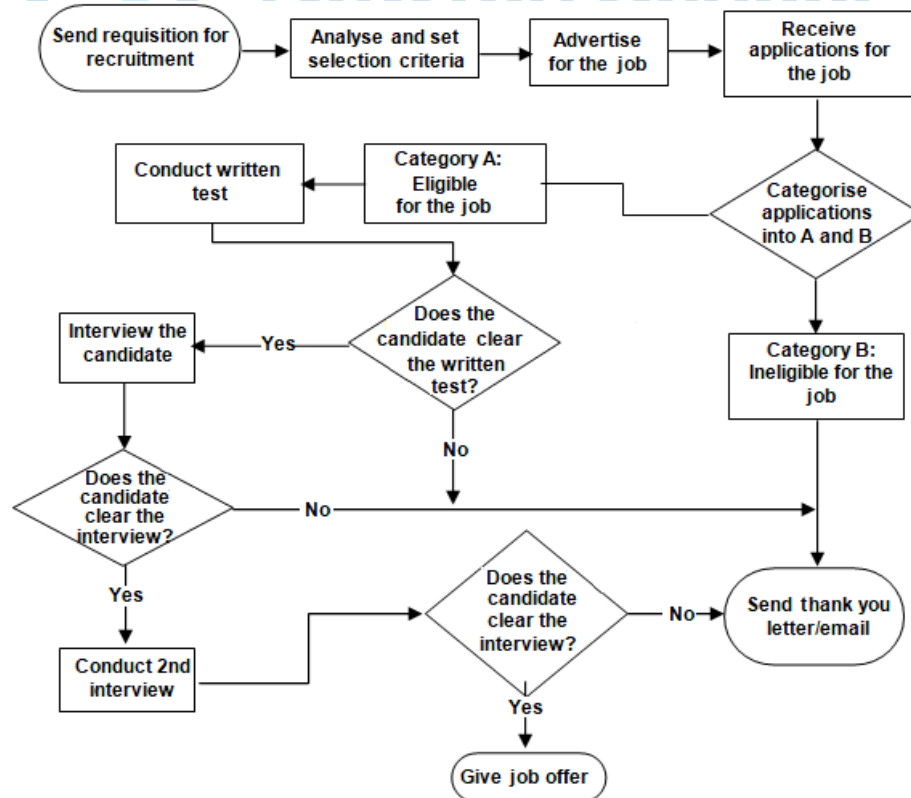
19. In a group of 70 participants, 25 enjoy painting, 30 enjoy photography, and 20 enjoy writing. Among them, 12 enjoy both painting and photography, 8 enjoy photography and writing, and 5 enjoy both painting and writing. If 15 participants do not engage in any of these artistic activities, how many participants are involved in exactly two of these artistic pursuits?
- (1) 15 (2) 12 (3) 10 (4) 8 (5) 5

20. In a city with a population of 10,000 people, 70 percent of the residents enjoy playing sports, 45 percent are avid readers, and 15 percent do not engage in either sports or reading. How many residents are enthusiastic about sports but not interested in reading?
- (1) 3,000 (2) 4,000 (3) 4,500 (4) 5,000 (5) 5,500



Directions for questions 21 – 24:

Study the following diagram and answer the questions that follow in accordance to it.



21. According to the flowchart given above, how many different phases of assessment criteria does a candidate have to fulfil in order to bag a job offer?
- (1) 1 (2) 2 (3) 3 (4) 4 (5) None of these
22. Which of the following statements are true?
- A. All the candidates get a thank you letter/e-mail.
B. Even candidates who are unsuitable for the job do get a thank you letter/e-mail.
C. Even candidates who are rejected after assessment in phase 1 do get a thank you letter/e-mail.
D. Even candidates who are rejected after round 2 of interview do get a thank you letter/e-mail.
- (1) All A, B, C and D (2) B, C and D only (3) B and C only
(4) B and D only (5) C and D only
23. If 16 candidates apply for the job and 50% of the candidates get rejected at each stage, then what is the number of candidates who get the job?
- (1) 8 (2) 4 (3) 2 (4) 1 (5) Cannot be determined



24. How many times does a selection committee have to take a decision on a particular candidate?
(1) 1 time (2) 2 times (3) 3 times (4) 4 times (5) Cannot be determined

Directions for questions 25 – 27: Study the information given below carefully and answer the questions based on it.

A robot is placed in a maze represented by a rectangular grid. The robot can move forward, turn left, and turn right. Each cell in the grid represents either an open path or a wall. The robot starts at a specific location and is oriented in one of the cardinal directions (north, east, south, west). The goal is to navigate the robot through the maze to reach a specified destination. The robot is initially placed at cell co-ordinates (3, 3) in the maze, facing north. The maze is represented as a 5 x 5 grid, in which each cell is represented by a number where an even number denotes an open path, and an odd number denotes a wall. The robot can move one step at a time from its current cell.

Cell co-ordinates (a, b) imply a position represented by the ath column from the left and the bth row from the bottom.

5	8	6	28	65	32
4	97	99	62	43	42
3	6	28	14	66	72
2	36	57	55	53	38
1	58	56	54	74	98
	1	2	3	4	5

25. Which sequence of commands should the robot execute to reach the destination cell co-ordinates (3, 1)?
- (1) Turn right, move forward, move forward, turn left, move forward, move forward, turn right, move forward, move forward
 - (2) Move backward, move backward
 - (3) Turn right, move forward, move forward, turn right, move forward, move forward, turn right, move forward, move forward
 - (4) Turn left, move forward, move forward, turn right, move forward, move forward, turn left, move forward, move forward
 - (5) None of these
26. The robot is currently at position (2, 3) and facing east. If it executes the command "turn right" and then "turn left," what will be its new position?
- (1) (2, 4) (2) (3, 4) (3) (1, 3) (4) (2, 3) (5) None of these
27. By which of the following commands will the robot reach a perfect square number?



- (1) Turn left, turn left, move forward, turn left, move forward
- (2) Turn right, turn left, move forward, turn right, move forward
- (3) Turn right, move forward, turn right, move forward, turn left, move forward
- (4) Turn left, move forward, move forward, turn left, move forward
- (5) None of these

Directions for questions 28 – 30:

Study the information given below carefully and answer the questions based on it.

An AI-enabled driverless car is instructed to reach a desired place. There are eight types of instructions which are as follows:

Instruction A: East - 48 m

Instruction B: North-east - 40 m

Instruction C: West - 8 m

Instruction D: South-west - 40 m

Instruction E: North - 12 m

Instruction F: North-west - 40 m

Instruction G: South - 52 m

Instruction H: South-east - 40 m

North-east means 45 degrees north and 45 degrees east. The same applies for south-west, north-west and south-east.

28. If the instructions given to the car are ABCD, how far is it from its starting point?

- (1) 40 m (2) 48 m (3) 56 m (4) 72 m
- (5) None of these

29. Which of the following set of instructions will take the car the farthest?

- (1) DFB (2) FDH (3) BHB (4) BHD
- (5) None of these

30. Which of the following set of instructions will take the car to a position which is in purely North Direction from its starting point?

- (1) FEF (2) BAD (3) FDC
- (4) Either (1) or (3) (5) None of these

SECTION – 2: ENGLISH

Directions for questions 31 – 33: Read the following passage and answer the questions.

The 5G technology is knocking at our doors faster than what we expected. The advent of this next-generational technology is set to make radical changes in our lives. One of the potential beneficiaries of this 5G technology will be the healthcare sector. A research carried out by Massachusetts-based International Data Corporation lists out the possible changes in the health sector via 5G. The operation theatre is long modernised by introducing surgical robots. Robot assisted surgery is a lot safer method in which the surgeon operates on the



patient by giving commands to the surgical robot. Reinforcing surgical robots with 5G technology will help to operate them remotely, a crucial development to help patients during crisis situations when it's not possible to bring a doctor on the location, the report said. At present, a doctor must be present near the patient and control the robot from the room itself because the connectivity requires very low latency between command and execution. The pandemic has made us realise the need of a virtual platform to interact and avail treatment. This 5G technology can facilitate tele-treatment where doctors can smoothly treat patients from other corners of the country. It will allow the maintenance of social distancing norms and will also ensure quality healthcare in rural areas. The 5G technology will give better and stable video streaming quality necessary to closely look for the symptoms. Apart from this, the 5G supported tele-medicine system can also incorporate the patient records and large files of diagnostic imaging systems without straining the connectivity. There have also been innovations in the fields of AR and VR. Augmented Reality (AR) enhances your surroundings by adding digital features to a live view of the camera, whereas Virtual Reality (VR) is an entirely immersive experience that substitutes a virtual environment for the actual world.

Medical students can be taught using AR and VR technology without the need to perform the demonstration on a real corpse, the report states. The AR technology can also be used to project diagnostic imaging on a patient. These may increase the surgery precision by giving a less harmful surgical pathway alternative to the surgeon. The VR technology can be a ground-breaking application in mental health cases as it can help to reduce pain and anxiety. These technologies will get a boost when connected with 5G technology as it brings increased bandwidth and low latency. It will eliminate the motion sickness associated with VR usage by better syncing the user's movement with the visuals on the system. The low latency will also ensure that AR images are accurately placed during critical applications.

31. The primary purpose of the author of the passage is to
- (1) make a case for expedited adoption of 5G technology
 - (2) contrast efficiency of robotic and manual procedures
 - (3) underscore 5G's revolutionary potential in healthcare
 - (4) pin-point the suppressed development of medical sector
 - (5) None of these
32. Which of the following lines is an accurate example of 'personification' used by the author?
- (1) The operation theatre is long modernised by introducing surgical robots.
 - (2) The 5G technology is knocking at our doors faster than what we expected.
 - (3) The AR technology can also be used to project diagnostic imaging on a patient.
 - (4) The pandemic has made us realise the need of a virtual platform to interact and avail treatment.
 - (5) None of these
33. Which of the following features can be associated with the 5G technology?
- | | | |
|-----------------------------|----------------------------|---------------------|
| (1) Widespread availability | (2) Inexpensive operations | (3) Enhanced speeds |
| (4) Minimal equipment | (5) None of these | |

Directions for questions 34 – 36:

Read the following passage and answer the questions.



Champaran Satyagraha of 1917 was one of the most significant and impactful events in the Indian independence movement led by Mahatma Gandhi. It was a successful struggle against the British oppression and exploitation of the peasants in the Champaran district of Bihar.

The British had forced the peasants to cultivate indigo, a cash crop, on one-third of their lands, and had fixed the prices of indigo at a rate that was unfavourable to the peasants. The British landlords also forced the peasants to purchase the necessary supplies from the landlords' own shops, which were overpriced. The peasants were thus trapped in a cycle of debt and poverty, unable to improve their economic condition.

Gandhi, who had just returned to India from South Africa, was approached by Raj Kumar Shukla, a peasant from Champaran, who urged him to visit the district and help the peasants. Gandhi agreed and arrived in Champaran in April 1917.

Gandhi was arrested upon his arrival in Champaran, but this only served to increase the support for him among the peasants and the local people. Gandhi was eventually released, and he began his satyagraha, or nonviolent resistance, against the British oppression.

Gandhi organised a team of lawyers, including Rajendra Prasad, who later became the first President of India, to investigate the grievances of the peasants and gather evidence to support their case. The British authorities tried to intimidate and threaten the lawyers and the peasants, but Gandhi's team continued their work with courage and determination.

In June 1917, the British authorities agreed to set up an inquiry commission to investigate the grievances of the peasants. The commission found that the British landlords had indeed exploited the peasants, and recommended that the peasants be allowed to grow whatever they wanted on their lands, and that the prices of indigo be fixed at a fair rate.

The Champaran Satyagraha was a significant success for the Indian independence movement, and it marked the beginning of Gandhi's journey as a leader of the Indian masses.

The Champaran Satyagraha was not just a struggle against economic oppression, but also against racial discrimination and the British rule in India. Gandhi's message of nonviolent resistance, or satyagraha, inspired millions of Indians to join the independence movement and fight for their rights.

The Champaran Satyagraha was a turning point in the history of India, and it showed that the Indian masses had the power to challenge and overthrow the British rule. It was the first of many successful satyagrahas led by Gandhi, including the Salt Satyagraha and the Quit India Movement, that eventually led to India's independence in 1947.

34. What was the main grievance of the peasants in Champaran?
- | | |
|---|---|
| (1) Unfair pricing of their produce | (2) Increased tariffs on trading of crops |
| (3) Lack of access to water for farming | (4) Increased interest rates on farming loans |
| (5) None of these | |
35. All the following statements are correct, except:
- | |
|---|
| (1) The British authorities agreed to investigate the complaints of farmers from Champaran. |
| (2) The inquiry commission did not find any proof against the British landlords. |
| (3) Gandhi formed his own investigative team to look into the issue of Champaran farmers. |
| (4) The British authorities threatened the investigators and tried to suppress the investigation. |
| (5) Both 2 and 3 |



36. Which of the following statements best describes the impact of Gandhi's arrest in Champaran?
- (1) The locals were disappointed and disheartened by Gandhi's arrest, leading to a decrease in his support.
 - (2) The British authorities became more oppressive towards the locals, worsening their situation.
 - (3) The arrest of Gandhi heightened his popularity among the local population, ultimately leading to his release.
 - (4) The Satyagraha movement lost momentum and ultimately failed due to Gandhi's arrest.
 - (5) Only 3 and 4

Directions for questions 37 – 40:

Read the following passage and answer the questions.

Geopolitics, the study of international relations and the distribution of power among countries, is a complex and multifaceted field. Two key concepts in this field are soft power and hard power, which refer to different ways in which countries can exert influence over others.

Soft power, which is the ability to influence others through non-coercive means, such as cultural exports, diplomacy, and the promotion of values, has gained importance in recent years. Countries that can project a positive image of themselves and their values can attract other countries to their side and achieve their goals without resorting to military force in a benignant manner. For example, a country's diplomatic efforts, cultural exports such as movies and music, and humanitarian aid can all be forms of soft power.

However, while soft power can be effective in certain situations, it is not always sufficient to achieve a country's objectives. In some cases, hard power, which is the ability to influence others through coercive means such as military force or economic sanctions, may be necessary to ensure a country's security and protect its interests. For example, a country facing a threat from a hostile neighbour may need to use military force to defend itself, or use economic sanctions to pressure a country to change its behaviour.

Despite their differences, soft power and hard power are often used in combination in international relations. Countries may use soft power to build relationships with other countries and promote their values, while also exerting pressure by maintaining a vigorous military standing and using economic sanctions when necessary. This approach allows countries to achieve their objectives while projecting a positive image of themselves to the world.

The balance between soft power and hard power can vary depending on a country's goals and its place in the international system. For example, smaller countries may rely more on soft power to build relationships with larger countries, while larger countries may have more resources to devote to military and economic power. Similarly, countries that are part of a larger regional organisation may need to balance their own interests with the interests of the group as a whole.

In conclusion, soft power and hard power are two important aspects of geopolitics, and a combination of both can be an effective way for countries to achieve their objectives while projecting a positive image of themselves to the world. Understanding the balance between soft power and hard power is crucial for any country seeking to navigate the complex world of international relations.

37. Which of the following examples demonstrates the use of soft power in international relations?
- (1) Deploying armed forces to establish influence
 - (2) Imposing economic sanctions to pressurise a country
 - (3) Conducting diplomatic negotiations to resolve conflicts



- (4) Applying political intimidation to gain cooperation
(5) None of these
38. When might a nation resort to hard power to accomplish its goals?
- (1) When it needs to improve trade relationships with other nations
(2) When it seeks to increase the global acceptance of its values and beliefs
(3) When it encounters a danger from a belligerent adversary
(4) When it intends to support the humanitarian cause of another nation
(5) None of these
39. How might the distribution of soft power and hard power vary among different nations?
- (1) Less powerful nations may more often employ soft power to establish connections with more powerful nations.
(2) Countries with less military might may resort more to imposing economic sanctions.
(3) Nations with a stronger cultural presence may rely more on soft power to project their values.
(4) The balance between soft power and hard power is primarily determined by a country's economic strength.
(5) None of these
40. What is the primary benefit of adopting a comprehensive approach that involves both soft power and hard power in international relations?
- (1) It helps reduce the prevalence of cross-cultural conflicts and fosters harmony among nations.
(2) It ensures that a country's military always remains the primary means of achieving its objectives, even at the cost of projecting a negative image to the world.
(3) It reduces a country's vulnerability to political pressure and to economic sanctions by other nations.
(4) It allows a nation to accomplish its goals while concurrently portraying a favourable global image, consequently acquiring the collaboration of other nations.
(5) None of these

Directions: In the following question, two statements numbered I and II are given. There may be cause and effect relationship between the two statements. These two statements may be the effect of the same cause or independent causes. These statements may be independent causes without having any relationship. Read both the statements in the question and mark your answer accordingly.

41. **Statements:**

- I. In the past, pilots relied on old-school paper charts for getting weather and terrain updates.
II. Today, with improved connectivity, pilots can communicate better with on-the-ground air traffic control and flight operators and gain access to technologies that provide an accurate and intuitive view of the weather.
- (1) Statement I is the cause and statement II is its effect.
(2) Statement II is the cause and statement I is its effect.
(3) Both statements I and II are independent causes.
(4) Both statements I and II are effects of independent causes.



- (5) Both statements I and II are effects of some common cause.

Directions: In the following question, four statements have been provided. These statements form a coherent paragraph when properly arranged. Select the alternative representing the proper and logical sequencing of these statements.

42. A. With NITI Aayog's new definition, the anti-poverty programmes have to be redesigned to ensure generation of more employment, creation of assets, imparting productive skills and raising the incomes of the very poor.
- B. The anxiety for estimating the correct incidence of poverty is well appreciated, but did it call for approving the new methodology without a full-scale debate?
- C. The poverty estimates have always been a subject of debate and experts have differed on an agreed estimate of either the level of poverty or the number of persons below the poverty line, but there never has been such wide divergence in estimates.
- D. The issue requires a holistic approach.
- (1) ACDB (2) CABD (3) BDAC (4) BCAD (5) CDBA

Directions: Given below is a pair of events I and II. Read both events and decide the relationship between them. While giving your answer, assume that the given information is true.

43. **Event I:** The access and convenience for Indian students looking to pursue higher education in Australia has increased.

Event II: India has been upgraded in the student visa evaluation framework.

- (1) I is an effect, but II is not its immediate and principal cause.
- (2) I is the immediate and principal cause and II is the effect.
- (3) I is an effect and II is its immediate and principal cause.
- (4) II is an effect, but I is not its immediate and principal cause.
- (5) None of these

Directions: Read the following text and answer the question.

The strength of weak ties is really a cornerstone of social science. In a 1973 study, researchers interviewed people late in their career and asked them about their experiences with job changes. Before this groundbreaking research, many had assumed that new positions came from sources such as close personal friends who would put in a good word, head-hunters who would seek out strong candidates or public advertisements. But the study analysis showed that people _____.

44. Which choice most effectively completes the paragraph?
- (1) who belong to different communities can offer a whole new set of information and be helpful acting as a bridge and providing new opportunities
- (2) who make weak ties have some unobserved characteristics that also make them more productive, have good ideas and get better jobs, promotions and wages



- (3) associated concepts to explain a wide range of phenomena, but the most useful ties were the ones that they did not interact with very often
- (4) actually got new jobs most frequently through friends of friends often someone the job seeker had not known before they started looking for a new position
- (5) None of these

Directions: Read the following text and answer the question.

Translator of Japanese literature Sam Bett once said that translating is the slowest form of reading. When translating, _____ each word has on the whole text and, no matter how fast of a reader you are, there's really no way around the time you'll need to do such meticulous work. As translators, we have to slow down our reading and feed the text into the sausage machine of our target language generator—a machine that science doesn't quite seem to understand yet—and hope the sausage arrives before our publishers' deadline.

45. Which choice completes the text so that it conforms to the conventions of Standard English?
- (1) your consideration of the impact of (2) the need of impact consideration
- (3) the impact of the consideration need (4) you need to consider the impact
- (5) None of these

Directions: In the following question, four statements have been provided. These statements form a coherent paragraph when properly arranged. Select the alternative representing the proper and logical sequencing of these statements.

46. A. The repeal in 1770 of the Townshend duties (except for that on tea, maintained to uphold the principle of Parliamentary taxation) led to a relative lull in colonial agitation for a few years - though even this lull was disturbed by the "Boston Massacre" and other incidents.
- B. Universal resistance to the acceptance of this tea was dramatised by the Boston Tea Party, which provoked the Boston Port Act and the other 'Intolerable Acts' of the spring of 1774.
- C. However, the attempt of Parliament to bail the vast East India Company out of its financial difficulties, by giving its tea a privileged position in the American market, reopened old issues that had been set aside but never resolved.
- D. The crisis now entered a new phase: now the colonists felt their rights and liberties were being invaded in many ways.
- (1) ACBD (2) BCDA (3) ADBC (4) BDAC (5) DCBA

Directions: Some alternative summaries are given below the text. Choose the option that best captures the essence of the text.

47. The pattern is very clear: the young countries are poor, and the old countries are rich. So why do people fear population ageing? The analogy to individual ageing suggests that as populations get older, they grow frail and lose mental acuity. There is also an indicator, called the dependency ratio, at work; it assumes that every adult below age 65 contributes to society and everybody above 65 is a burden. And the proportion of people older than 65 is bound to rise.



- (1) Popular perceptions and assumptions have caused nations witnessing an increasing share of aged population to become fearful.
- (2) Since nations with a majority of aged people are rich, the fear that aged are a liability is irrational.
- (3) People's fear of an ageing population is grounded in the myth that aged people are worthless.
- (4) The fear that an ageing population, regardless of its economic soundness, is a liability to a nation is not unfounded.
- (5) None of these

Directions: Read the following text and answer the question that follows.

Recruitment agent: Job seekers do not possess as good writing skills as they used to. Almost all of the covering letters that I have got this year have been poorly written and ungrammatical.

48. Which of the following is the most serious weakness in the argument made by the recruitment agent?
- (1) It requires corroboration that the change in a particular agent's perspective is illustrative of a similar change among job seekers in general.
 - (2) It offers no evidence of the fact that the recruitment agent is an accurate assessor of writing skills.
 - (3) It does not take into consideration the possibility that the agent is a poor recruiter.
 - (4) It fails to present any evidence to the contrary.
 - (5) It does not take into consideration the possibility that the agents have started ignoring grammatical mistakes.

Directions: Read the following text and answer the question.

How did chilli peppers become part of the human diet an estimated 6,000 to 10,000 years ago? And why were they eventually embraced by the rest of the world? Some evolutionary biologists have proposed that the human propensity for spiciness is borne of necessity. They posit that because peppers (and other spicy foods, including wasabi) have natural antimicrobial properties that can help preserve perishable foods, humans developed a taste for them, particularly _____.

49. Which choice most effectively completes the paragraph?
- (1) when their natural antimicrobial properties can help preserve perishable foods
 - (2) in tropical climates where food spoils easily
 - (3) when they are known to have a natural ability to ward off fungi
 - (4) as a means to distinguish themselves from other cultural groups
 - (5) None of these

Directions: Read the following text and answer the question.

Spiciness is a burning sensation caused by capsaicin in food. When we eat spicy food, capsaicin stimulates receptors in our mouth called TRPV1 receptors and triggers a reaction. The purpose of TRPV1 receptors is thermoreception — the detection of heat. This means they are supposed to _____ us from consuming food that burns.



50. Which choice completes the text with the most logical and precise word?
(1) protect (2) separate (3) deter (4) refrain (5) None of these

SECTION – 3: MATHEMATICS

51. Which of the following is the largest three-digit number that is exactly divisible by 3, 4 and 6 but not by 83?

(1) 999 (2) 996 (3) 984 (4) 972 (5) 913

52. The Blooming Garden Company is preparing flower bouquets for an event. They have 92 roses, 460 tulips, 184 daisies, 207 lilies, and 552 sunflowers. The flowers need to be arranged into bouquets in such a way that each bouquet contains only one type of flower, and all the bouquets must have an equal number of flowers.

What is the minimum number of bouquets required to arrange all the flowers?

(1) 23 (2) 46 (3) 63 (4) 65 (5) None of these

53. The remainder obtained when $16^{203} + 11^{203}$ is divided by 18 is

(1) 0 (2) 1 (3) 2 (4) 5 (5) 9

54. If $365xy$ is divisible by 30, then the least value of $x + y =$

(1) 9 (2) 5 (3) 0 (4) 1 (5) 4

55. Xavier, Yacub, and Zeenat launch a joint online business. Xavier makes a Rs. 3 lakh investment, Yacub a Rs. 4.5 lakh investment, and Zeenat a Rs. 6 lakh investment. The business produces a 18 lakh rupee profit in its first year. Yacub and Zeenat split a portion of the profit in a ratio of 1 : 3, and the remaining profit is distributed among Xavier, Yacub, and Zeenat in proportion to their initial contributions. The sum that Zeenat receives is $\frac{3}{2}$ of Yacub's sum. What sum of money does Yacub receive?

(1) Rs. 6,80,467.16 (2) Rs. 5,80,645.16 (3) Rs. 9,75,005.16
(4) Rs. 4,87,257.16 (5) None of these

56. In a competition, Jai is 160% as efficient as Amit, who is 250% as efficient as Birbal, who is 300% as efficient as Chetan, who is 600% as efficient as Daya. If Daya takes 36 days to complete a piece of work and gets a share of Rs. 5,000 out of the total allotted money, find the total allotted money in rupees.

(1) 2,50,000 (2) 5,40,000 (3) 7,10,000 (4) 7,00,000
(5) None of these

57. In a company, there are two teams, team A and team B. The number of employees in team A is 35, while the number of employees in team B is 50. The average salary of all the employees in the company is Rs. 60,000. If the average salary of the employees in team A is 30% higher than that of employees in team B, then what is the approximate average salary of employees in team B?

(1) Rs. 50,000 (2) Rs. 56,200 (3) Rs. 53,400 (4) Rs. 51,200 (5) None of these



58. The salary of Priya is 90% more than the salary of Neha. The monthly expenditure of Neha is 25% more than that of Ritu. If the monthly expenditure of Ritu is one-third of Neha's salary, then how much percent (approximately) is the total expenditure of Ritu and Neha less than Priya's salary?

- (1) 16.84% (2) 60.53% (3) 36.84% (4) 46.84%
(5) None of these

59. $y = [(x - 2)^2]^{1/2}$ and $y = (3x + 22)^{1/2}$

Find the least value of x , which satisfies the above equations.

- (1) -15 (2) -11 (3) -7 (4) -2
(5) None of these

60. How many positive integer pairs (x, y) satisfy the inequality $4^{x+y} - 10(2^{x+y}) + 64 \leq 10(2^{x+y})$?

- (1) 1 (2) 2 (3) 3 (4) 6 (5) 4

61. A group of six students plans to raise at least Rs. 480 for a charity event. However, the amount raised by each student is subject to a condition: the amount raised by each student must be a prime number. Five of the students have already raised Rs. 37, Rs. 59, Rs. 71, Rs. 83, and Rs. 97, respectively. What is the minimum amount that the sixth student must raise to ensure that the group achieves their goal of raising at least Rs. 480?

- (1) 107 (2) 139 (3) 133 (4) 137 (5) None of these

62. P, Q, R, S, T and U are consecutive multiples of a number ' x '. Find the difference between the sum of the three largest and three smallest multiples.

- (1) $8x$ (2) $9x$ (3) $10x$ (4) $11x$ (5) $12x$

63. A runner started a race at 9 a.m. and ran at a speed of 12 km/h. The runner rested for 20 minutes and then ran back to the starting point at a speed of 18 km/h. The runner reached the starting point at 11:05 a.m. Calculate the total distance covered by the runner.

- (1) 63 km (2) 13 km (3) 12.6 km (4) 25.2 km (5) None of these

Directions: In this problem, a question is given followed by three statements numbered A, B and C. Read the statements carefully and decide which of them is/are sufficient/required to answer the question.

64. What is the time taken by Lovely to complete a 10-kilometre race?

Statements:

- A. If Lovely increases her speed by 20%, she will finish the race 5 minutes earlier.
B. If Lovely decreases her speed by 15%, she will finish the race 6 minutes later.
C. Lovely's speed is 8 km/hr.

- (1) Only C is sufficient. (2) Only B and C are sufficient.
(3) Only A and C are sufficient. (4) Each of the statements A, B and C alone is sufficient.
(5) None of the statements A, B and C is sufficient.



65. Jatin and Sonia are driving towards each other on a straight road. Jatin drives at a constant speed of 80 kilometres per hour, while Sonia drives at a constant speed of 60 kilometres per hour. They start from locations that are 200 kilometres apart. A bird sitting on Jatin's car starts flying towards Sonia's car once they start driving and continues to fly back and forth between the cars until they meet. If the bird flies at a constant speed of 90 kilometres per hour, how far does the bird fly before Jatin and Sonia meet?
- (1) 128.7 kilometres (2) 148.7 kilometres (3) 168.7 kilometres
(4) 228.7 kilometres (5) None of these
66. Mr. Ganesh bought 4,500 shares of ABC company at the rate of Rs. 14 each. He sold 3,000 shares at a cost of Rs. 18 each, and then after one week, the price of the share decreased by one-third of its value and he sold the remaining shares. Find the profit or loss made by Mr. Ganesh in the entire transaction considering brokerage charged is negligible.
- (1) Rs. 4,500 (2) Rs. 6,000 (3) Rs. 9,000 (4) Rs. 10,000 (5) Rs. 12,000
67. A jewellery store owner labels the prices of his products at 20% above the cost price. He sells 70% of the jewellery at 10% discount, and the remaining at 20% discount on the marked price. What is the overall profit/loss percentage for the jewellery store owner?
- (1) 3.6 (2) 4.4 (3) 5.6 (4) 8.2 (5) None of these
68. Rohan purchased a rare antique watch at successive discounts of 35% and 20%. He invests 25% of the cost price in restoring the watch and decides to sell it at a profit of 50%. However, due to high demand, he receives an offer from a collector to buy the watch at a price that is 10% higher than his intended selling price. If the printed price of the watch is Rs. 2,00,000, then what is the final sale price of the watch?
- (1) Rs. 2,04,500 (2) Rs. 2,18,500 (3) Rs. 2,14,500 (4) Rs. 2,00,500 (5) None of these
69. Dinesh, a farmer, deposited Rs. 64,000 in a State Farm Bank deposit scheme for two years and got a simple interest of Rs. 16,000. He withdrew the entire amount after 2 years including interest and deposited in Kanara Bank's saving scheme which offered 2.5% extra interest as compared to State Farm Bank and that too compounded annually. Calculate the total compound interest obtained by Dinesh after 2 years.
- (1) Rs. 26,400 (2) Rs. 25,800 (3) Rs. 24,000 (4) Rs. 22,600 (5) Rs. 20,800
70. Brijesh and Varun borrowed same amount from ABC Finance at the same rate. Brijesh borrowed the money for 2 years and he had to pay Rs. 4,160 including compound interest compounded annually. Varun borrowed the money for 1 year and he had to pay Rs. 4,000 including simple interest. At what rate percent did they borrow the money?
- (1) 4% (2) 8% (3) 12% (4) 14% (5) 15%
71. In a survey conducted among students at a university, it was found that 80% of the students like reading books, and 60% of the students like watching movies. If at least $y\%$ of the students enjoy both reading books and watching movies, then
- (1) $y > 60$ (2) $0 \leq y \leq 40$ (3) $40 \leq y \leq 60$ (4) $40 \leq y \leq 80$ (5) None of these



72. If M is the maximum values of the function $f(x) = \frac{3x^2 - 6x + 12}{x^2 + 2x + 4}$, then the value of M is:
- (1) 2 (2) 5 (3) 9 (4) 11 (5) 18

73. In a group of 13 people, 4 are quadruplets (looking exactly identical), while the others are unique individuals. If you randomly select 4 persons from the group, how many different combinations of individuals can you form?

(1) 56 (2) 123 (3) 174 (4) 256 (5) None of these

74. For $n \geq 1$, the sum of n terms of a sequence is given as $\sum_{i=1}^n (-1)^{i+1} x_i = n^3 + 3n^2 - 2n$. What will be the sum of the 98th and 99th terms of the sequence?

(1) -594 (2) 594 (3) 2 (4) -2 (5) -4

75. Consider the given equation.

$$\sin^{4\alpha+2} \int_{2\alpha}^{4\alpha} \cos 4x dx = 0$$

Which of the following is not the value of α in the interval $\left[-\frac{\pi}{4}, 0\right]$?

(1) $\alpha = -\frac{\pi}{4}$ (2) $\alpha = -\frac{\pi}{12}$ (3) $\alpha = 0$ (4) $\alpha = -\frac{\pi}{6}$

- (5) None of these

76. If $f(x) = \sin(a \log x)$ and $y = f\left(\frac{2x-3}{3x+4}\right)$, then $\frac{dy}{dx}$ at $x = 2$ is equal to

(1) $\frac{a \cos(a \log 10)}{10}$ (2) $\frac{17a \cos(a \log 10)}{10}$ (3)

$\frac{17a \cos(a \log 10)}{10}$

(4) $\frac{17a \cos(\log 10)}{10}$ (5) $\frac{10a \cos(a \log 10)}{17}$

77. If $\log_{x^2+3x+2}(x^2+5x+6) = y$, for all $x > 0$, then $\left[\frac{dy}{dx}\right]_{at x=1} =$



(1) $-\frac{\log 73728}{12[\log 6]}$

(2) $\frac{\log 221184}{12[\log 6]^2}$

(3) $-\frac{\log 221184}{12[\log 6]^2}$

(4) 1

(5) None of these

78. Find the absolute value of the determinant for a 4×4 matrix having all its entries of +2's and -2's such that the sum of each row and column of the matrix is 4.

(1) 0

(2) 2

(3) 4

(4) -2

(5) None of these

79. A matrix P is given as $\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where $x > y > z > 0$. Then the determinant of P will always be

(1) positive

(2) negative

(3) 0

(4) Either 1 or 2

(5) None of these

80. An individual has established a reputation for being untruthful, with an average tendency to fabricate falsehoods in 3 out of 7 instances. In a given scenario, this person rolls two dice and asserts that the sum of the outcomes on both dice exceeds 10. What is the probability that this claim indeed corresponds to an actual sum greater than 10?

(1) $\frac{3}{7}$

(2) $\frac{11}{28}$

(3) $\frac{4}{37}$

(4) $\frac{17}{28}$

(5) None of the these



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NSAT MOCK – 1

ANSWER KEY

1. (1) 2. (1) 3. (1) 4. (1) 5. (1) 6. (4) 7. (2)
 8. (4)
 9. (2) 10. (2) 11. (4) 12. (2) 13. (2) 14. (2) 15. (2)
 16. (3)
 17. (3) 18. (3) 19. (3) 20. (2) 21. (4) 22. (2) 23. (4)
 24. (5)
 25. (3) 26. (4) 27. (4) 28. (1) 29. (3) 30. (1) 31. (3)
 32. (2)
 33. (3) 34. (1) 35. (2) 36. (3) 37. (3) 38. (3) 39. (1)
 40. (4)
 41. (5) 42. (2) 43. (3) 44. (4) 45. (4) 46. (1) 47. (4)
 48. (1)
 49. (2) 50. (3) 51. (3) 52. (4) 53. (5) 54. (4) 55. (2)
 56. (3)
 57. (3) 58. (2) 59. (4) 60. (4) 61. (4) 62. (2) 63. (4)
 64. (4)
 65. (1) 66. (3) 67. (2) 68. (3) 69. (2) 70. (1) 71. (3)
 72. (3)
 73. (4) 74. (2) 75. (4) 76. (2) 77. (3) 78. (1) 79. (2)
 80. (3)

EXPLANATIONS

1. **Answer:** (1)

All scores were in whole numbers. None of them scored 100% marks. Only C scored less than 50% marks. This means C scored least marks and maximum marks is 9.

B scored 4 marks more than A. This is possible only when B scored 9 marks and A scored 5 marks.

D scored 7 marks. F scored more marks than E. This is possible only when E scored 6 marks and F scored 8 marks. Therefore, Order of marks is $B > F > D > E > A > C$.

Thus, B scored the maximum marks.

2. **Answer:** (1)

Math Club	Debate Team	Chess Club	Cheerleader
Anjali	Simran	Lily	Inder
Yasmin	Yasmin		Anjali
Simran			Yasmin
Jenny			

So, out of the given options, it is Yasmin who is both in the debate team and a cheerleader.



Hence, option 1 is correct.

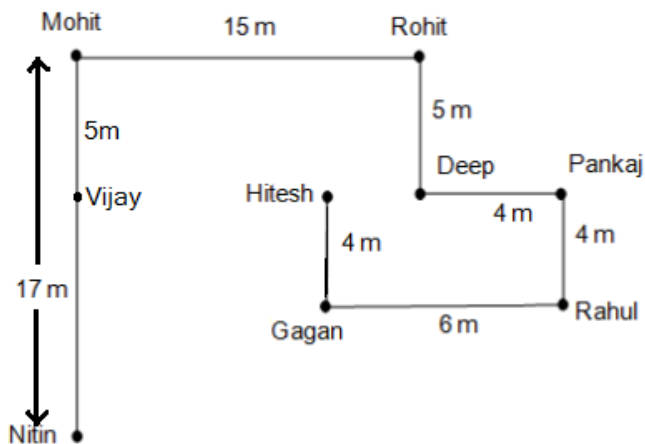
3. **Answer:** (1)

Math Club	Debate Team	Chess Club	Cheerleader
Anjali	Simran	Lily	Inder
Yasmin	Yasmin		Anjali
Simran			Yasmin
Jenny			

So, out of the given options, it is Yasmin who is in the math club, debate team and is a cheerleader.
Hence, option 1 is correct.

4. **Answer:** (1)

According to the question,



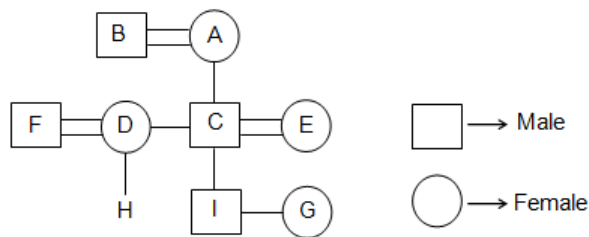
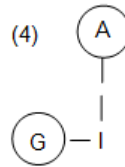
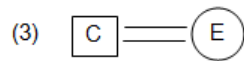
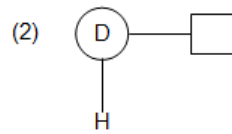
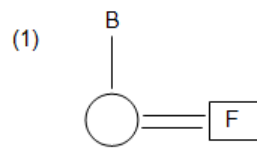
Distance between Pankaj and Vijay = 15 m + 4 m = 19 m

5. **Answer:** (1)

In the evening, the sun is in the west direction and the shadow will fall in the east. Now she could see the reflection of the sun in the mirror, meaning she was facing her back towards the sun and she was facing east. The traffic was diverted towards her left, which is the north direction.

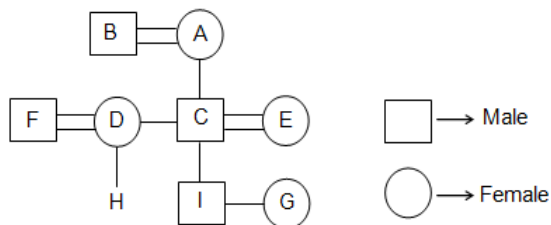
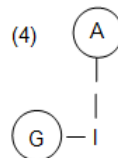
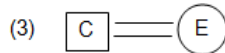
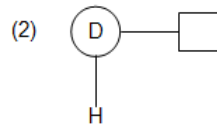
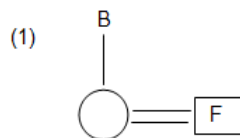


6. **Answer:** (4)



Gender of H is not clear. Therefore, option 4 is the correct answer.

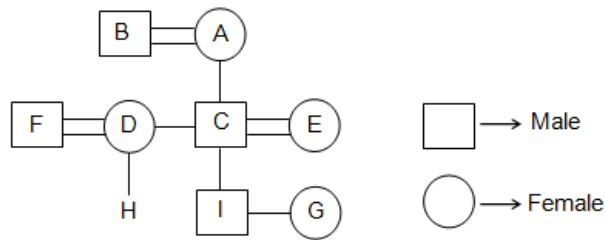
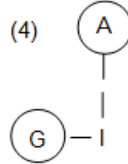
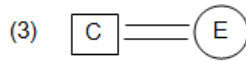
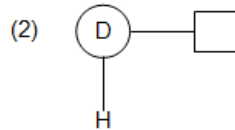
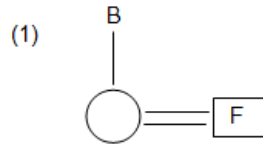
7. **Answer:** (2)



F is the brother-in-law of E's husband.



8. **Answer:** (4)



B is the father-in-law of E.

9. **Answer:** (2)

According to the given code, the words are arranged according to the alphabetical order or the way in which will appear in the dictionary. So, 'Never Let The Fear Of Striking Out Keep You From Playing The Game' will be written as 'Fear From Game Keep Let Never Of Out Playing Striking The The You'.

10. **Answer:** (2)

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Alphabetic position values: 5-4-21-3-1-20-5-4

Now, $E + U = 5 + 21 = 26$

$D + C = 4 + 3 = 7$

$U + A = 21 + 1 = 22$

$C + T = 3 + 20 = 23$

$A + E = 1 + 5 = 6$

$T + D = 20 + 4 = 24$

Similarly, the code of DISCOVER will be:

$D + S = 4 + 19 = 23$

$I + C = 9 + 3 = 12$

$S + O = 19 + 15 = 34$

$C + V = 3 + 22 = 25$

$O + E = 15 + 5 = 20$

$V + R = 22 + 18 = 40$

Now, the code of AIRPORT will be:

$A + R = 1 + 18 = 19$



$$I + P = 9 + 16 = 25$$

$$R + O = 18 + 15 = 33$$

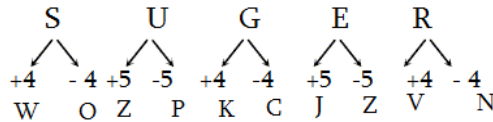
$$P + R = 16 + 18 = 34$$

$$O + T = 15 + 20 = 35$$

Therefore, AIRPORT will be coded as [1925333435].

11. **Answer:** (4)

The first letter of SUGER is decoded as +4 and -4, the second letter as +5 and -5, then again the third letter as +4 and -4, and so on.



Following the same logic, we get that QITJQIRHCU is the code of MOMMY.

12. **Answer:** (2)

Let the total unsold bags on Tuesday = x

Number of bags manufactured by B on Tuesday = $180 - 30 + x$

Number of bags manufactured by B on Wednesday = $200 + 20 - x$

$$ATQ, (200 + 20 - x) - (180 - 30 + x) = 10$$

$$220 - x - 150 - x = 10$$

$$70 - 2x = 10$$

$$2x = 60$$

$$x = 30$$

$$\text{Required ratio} = 30 : 20$$

$$= 3 : 2$$

Hence, answer option (2) is correct.

13. **Answer:** (2)

Let the total bags unsold by A on Tuesday = a

Let the total bags sold by A on Wednesday = b

$$\text{Now, } b = a + 190 \dots (i)$$

Let the total number of bags manufactured by A on Tuesday = T

$$\text{So, } T + 20 = 150 + a$$

$$\text{Or } T = 150 + a - 20 = 130 + a \dots (ii)$$

Let the total number of bags manufactured by A on Wednesday = W

$$\text{So, } W + a = b + 50 \dots (iii)$$

Plugging in the value of b from equation (i) into equation (iii), we get $W + a = a + 190 + 50$

$$\text{Or } W = 240 \dots (iv)$$

$$\text{Now, } a + 50 = 0.2(T + W) \dots (v)$$

On plugging in the value of W as 240 from equation (iv) into equation (v) and that of T as $130 + a$ from equation (ii) into equation (v), we get $a + 50 = 0.2(130 + a + 240) \dots (vi)$

$$\text{Or } a = 30 \dots (vii)$$



Now, $T + W = 130 + a + 240 = 130 + 30 + 240 = 400$

Thus, number of bags manufactured by A on Tuesday and Wednesday = 400

Hence, answer option (2) is correct.

14. **Answer:** (2)

Average of the bags sold by A, B and C on Wednesday = 180

Total bags sold by A, B and C on Wednesday = 180×3

= 540

Number of bags sold by A on Wednesday = $540 - (200 + 180)$

= 160

Let the number of unsold bags of A = x

Then, number of unsold bags of B = $2x$

Average of the bags manufactured by A, B and C on Wednesday = 170

Total bags manufactured by A, B and C on Wednesday = 170×3

= 510

Number of bags manufactured by A on Wednesday = $160 + 50 - x$

= $210 - x$

Number of bags manufactured by B on Wednesday = $200 + 20 - 2x$

= $220 - 2x$

Number of bags manufactured by C on Wednesday = $180 + 20 - 30$

= 170

ATQ, $210 - x + 220 - 2x + 170 = 510$

$600 - 3x = 510$

$3x = 90$

$x = 30$

Therefore, number of unsold bags of A on Tuesday = 30

Number of unsold bags of B on Tuesday = 60

Required ratio = $(30 + 60 + 30) : (50 + 20 + 20)$

= 120 : 90

= 4 : 3

15. **Answer:** (2)

Cost Price of item D = Rs. 1,200 [given]

From the graph,

Mark-up percentage = 50% and discount percentage = 28%

By using the given formula,

Marked Price = Cost Price $\times (1 + \text{Mark-up}\%)$

Marked Price of item D = $1,200 \times (1 + 0.50) = \text{Rs. } 1,800$

Now, Selling Price = Marked Price - (Discount% $\times$ Marked Price) [given]

Selling Price of item D = $1,800 - (0.28 \times 1,800) = \text{Rs. } 1,296$

Profit = Selling Price - Cost Price = $1,296 - 1,200 = \text{Rs. } 96$

Profit Percentage = $(\text{Profit} / \text{Cost Price}) \times 100 = \frac{96}{1,200} \times 100 = 8\%$



Therefore, the profit percentage for item D is 8%.

16. **Answer:** (3)

Selling Price of item C = Rs. 510

Let its marked up price be x.

From the bar graph,

Discount percentage of C is 15%.

By using: Selling Price = Marked Price – Discount (in Rs.)

$$510 = x - 0.15 \times x$$

$$510 = 0.85x$$

Marked Price of item C = Rs. 600

Further,

Selling Price of item D = Rs. 1,224

Here, let the Marked Price of item D be y.

From the bar graph,

Discount percentage of C is 28%.

Similarly,

$$1,224 = 0.72 \times y$$

$$\text{Marked Price of D} = 1,224 / 0.72 = \text{Rs. } 1,700$$

$$\text{Required ratio} = x/y = 600 : 1,700 = 6 : 17$$

17. **Answer:** (3)

From the graph,

Initial discount = 26% and additional discount = 13%

Let the marked price of item E be x.

So, ATQ

$$(100 - 13)\% \text{ of } (100 - 26)\% \text{ of } x = \text{Rs. } 450.66$$

$$\Rightarrow 0.74 \times 0.87 x = \text{Rs. } 450.66$$

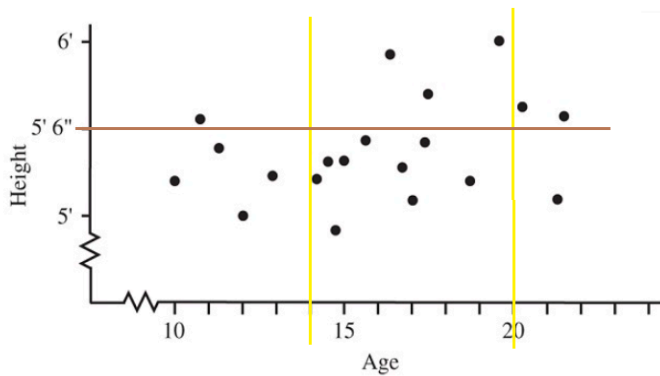
$$\Rightarrow 0.6438 x = \text{Rs. } 450.66$$

$$\Rightarrow x = \text{Rs. } 700$$

Hence, marked price of item E = Rs. 700

18. **Answer:** (3)

Total number of selected students = 20



From the above graph,

There are 9 dots which satisfy both the given conditions of below 5' 6" and are between 14 and 20 years.

$$\text{Required percentage} = \frac{9}{20} \times 100 = 45\%$$

19. **Answer:** (3)

Total participants = 70

Participants who enjoy painting = $n(P) = 25$

Participants who enjoy photography = $n(Ph) = 30$

Participants who enjoy writing = $n(W) = 20$

Participants who enjoy both painting and photography = $n(P \cap Ph) = 12$

Participants who enjoy photography and writing = $n(Ph \cap W) = 8$

Participants who enjoy both painting and writing = $n(P \cap W) = 5$

Participants who do not engage in any of these activities = 15

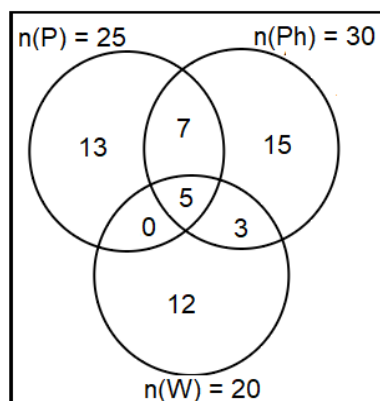
Therefore, participants who engage in all of these activities = $70 - 15 = 55$

Now,

$$n(P \cup Ph \cup W) = n(P) + n(Ph) + n(W) - n(P \cap Ph) - n(Ph \cap W) - n(P \cap W) + n(P \cap Ph \cap W)$$

Substituting the values, we get

$$n(P \cap Ph \cap W) = -25 - 30 - 20 + 12 + 8 + 5 + 55 = 5$$



Number of participants who are involved in exactly two of these activities = $7 + 0 + 3 = 10$

20. **Answer:** (2)

Total population in the city = 10,000

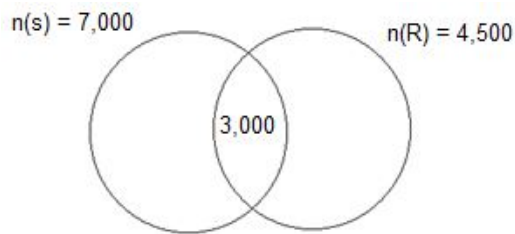
Number of residents who like playing sports = $0.7 (10,000) = 7,000$

Number of residents who like reading books = $0.45 (10,000) = 4,500$



Number of residents who do not engage in either sports or reading = 15% of the total population = 1,500

Number of residents who like sports or reading = $10,000 - 1,500 = 8,500$



Number of readers who like both sports and reading books = $7,000 + 4,500 - 8,500 = 3,000$

So, number of residents who have interest in sports but are not interested in reading = $7,000 - 3,000 = 4,000$



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21. **Answer:** (4)

According to the flowchart, a candidate has to pass through 4 selection criteria to bag a job offer. These are:

1. Application being matched with the application criteria
2. Assessment test matching the assessment test criteria
3. Clearing the round 1 of the interview
4. Clearing the round 2 of the interview

Thus, answer option 4 is correct.

22. **Answer:** (2)

Candidates who get the final job offer are not sent a thank you letter/e-mail. Rest of the candidates do get it irrespective of the phase/stage they have been rejected at. Thus, answer option 2 is correct.

23. **Answer:** (4)

At the application stage itself, 8 candidates get rejected.

Then, in the written examination, 4 candidates get rejected.

At the first interview, 2 candidates get rejected, and then at the second interview, 1 of the 2 remaining candidates gets rejected.

Thus, only 1 candidate out of the 16 gets through.

Hence, answer option 4 is correct.

24. **Answer:** (5)

The information asked for cannot be deduced from the given flowchart. Hence, answer option 5 is correct.

25. **Answer:** (3)

The move backward command is not given in the instructions.

There are two paths for the robot to reach cell co-ordinates (3, 1), i.e. at number 54.

5	8	6	28	65	32
4	97	99	62	43	42
3	6	28	14	66	72
2	36	57	55	53	38
1	58	56	54	74	98
	1	2	3	4	5

26. **Answer:** (4)

Since the robot is only turning, not moving forward, it will remain in the same cell.

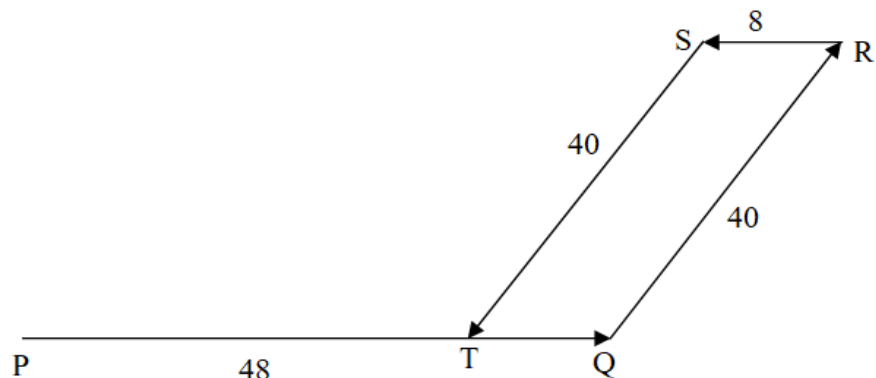


5	8	6	28	65	32
4	97	99	62	43	42
3	6	28	14	66	72
2	36	57	55	53	38
1	58	56	54	74	98
	1	2	3	4	5

27. **Answer: (4)**
36 is a perfect square number.

5	8	6	28	65	32
4	97	99	62	43	42
3	6	28	14	66	72
2	36	57	55	53	38
1	58	56	54	74	98
	1	2	3	4	5

28. **Answer: (1)**
According to the instructions, the driverless car traverses the path PQRST as below:



$$PT = 48 - 8 = 40 \text{ m}$$

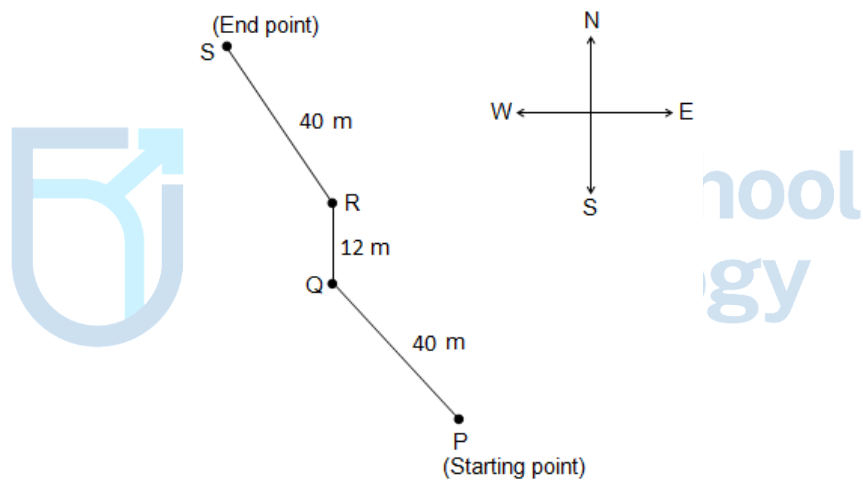
29. **Answer: (3)**
The instructions B and D, and also F and H are opposite.
Therefore, there will be no change in the position due to these instructions. The actual distance is determined by the middle instruction alone.
In options 1, 2 and 4, the net change is only due to the middle instruction.



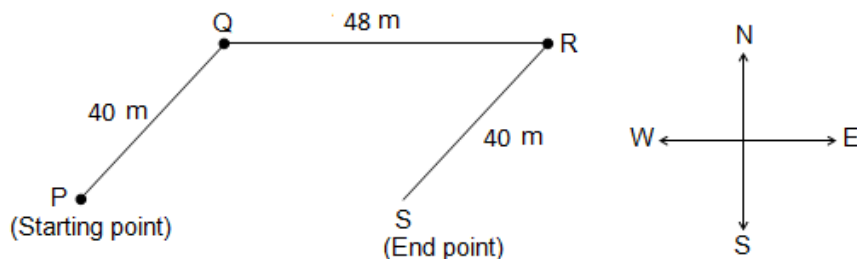
So, the net movement will be the highest for option 3.

30. **Answer:** (5)

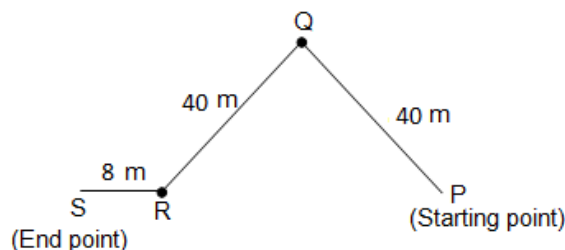
FEF



BAD



FDC



None of the given instructions takes the car to a position which is North from its starting point. Hence, option 5 is the correct answer.

31. **Answer:** (3)

The author is primarily trying to highlight the importance of 5G technology in the healthcare sector. The author lists how the reach, efficiency and availability of medical facilities would improve once 5G technology comes into operation. It will also give a boost to AR and VR technologies and their applications in the medical field. (1) is incorrect as although it is a favourable scenario, the author does not primarily focus on expedited adoption of such technology. Option 3 is the answer.

32. **Answer:** (2)



'Personification' means the act of giving a human quality or characteristic to something which is not human, or something which is abstract. Here, the 5G technology is personified by attaching a characteristic 'knocking at our doors'. Thus, option 2 is the answer.

33. **Answer:** (3)

Except 3, none of the features can be associated with the 5G technology. Option 3 can be inferred from the benefits like better and stable video streaming, low latency and increased bandwidth. We cannot infer that the 5G technology will be widely accessible and at cheaper costs. Also, nothing about the operations of the 5G technology are highlighted in the passage.

Thus, option 3 is the answer.

34. **Answer:** (1)

The main grievance of the peasants in Champaran was the forced cultivation of indigo on their lands by the British landlords. The British government had established a system of indigo cultivation that required farmers to grow indigo on 1/3rd of their land, regardless of the type of soil, water availability, or the crops the farmers preferred to grow. The farmers were also paid very little for their indigo, which further exacerbated their economic hardships.

35. **Answer:** (2)

The other three statements are correct as per the given text. The British authorities did agree to investigate the complaints of farmers from Champaran and the inquiry commission did find evidence of exploitation of peasants. Gandhi did form his own investigative team to look into the issue of Champaran farmers (and the British authorities did threaten the investigators and tried to suppress the investigation. Refer to, 'Gandhi organised a team of lawyers ... to investigate the grievances of the peasants ... British authorities tried to intimidate and threaten the lawyers ... In June 1917, the British authorities agreed to set up an inquiry commission to investigate the grievances of the peasants. The commission found that the British landlords had indeed exploited the peasants.'

36. **Answer:** (3)

Gandhi's arrest upon his arrival in Champaran served to increase the support for him among the peasants and the local people, as they saw him as a leader who was willing to stand up against the British oppression. The arrest of Gandhi was a significant event in the Champaran Satyagraha, as it helped to galvanise the support of the local people behind him. They organised protests and rallies demanding his release, which ultimately led to his release. Therefore, option 3 is the correct answer. Options 1, 2, and 4 are incorrect as they do not accurately describe the impact of Gandhi's arrest in Champaran.

37. **Answer:** (3)

Soft power refers to the use of non-coercive means to influence other countries, such as cultural exports, diplomacy, and the promotion of values. Diplomatic negotiations, which involve dialogue and compromise rather than the use of force or pressure, are an example of soft power. Military force and economic sanctions, on the other hand, are examples of hard power, which is the use of coercive means to influence others. Political pressure may also be considered a form of hard power, as it often involves threats or incentives to gain cooperation.



38. **Answer:** (3)

The question is asking when a country might need to use hard power, which is the ability to influence others through coercive means such as military force or economic sanctions, to achieve its objectives. Among the given options, only option 3 mentions the existence of a threat from a hostile neighbour, which may require the use of military force or economic sanctions to defend against or deter the adversary. Options 1 and 2 suggest using soft power to build relationships or promote values, while option 4 pertains to providing humanitarian aid, which is not directly related to the concept of hard power.

39. **Answer:** (1)

The question is asking how the balance between soft power and hard power might vary for different countries. Option 1 suggests that smaller countries may have fewer resources to devote to military and economic power, so they may rely more on soft power to build relationships with larger countries. Option 2 is incorrect as a country with less military strength is less likely to resort to economic sanctions. Option 3 suggests that a nation's cultural presence may influence its reliance on soft power, which is not necessarily true. Option 4 suggests that the balance between soft power and hard power is solely determined by a country's economic strength.

40. **Answer:** (4)

Using a combination of soft power and hard power enables a country to achieve its objectives while also building goodwill and trust among other nations. This approach helps a country to project a positive image of itself to the international community, which can foster cooperation and collaboration in international relations. Option 1 is incorrect because using soft and hard power does not guarantee the absence of conflicts, but can help mitigate them. Option 2 is incorrect because the approach involves using both soft and hard power, not just military power. Option 3 is incorrect because the use of hard power can also lead to political pressure and the imposition of economic sanctions by other nations.

41. **Answer:** (5)

In the past, pilots relied on old-school paper charts for weather and terrain updates. But why? It is not mentioned here. Probably it was due to the absence of technology. But now, pilots can experience improved connectivity. The reason behind this might be the progress in the technical arena. This implies that both the given statements are the effects of presence/absence of technology.

42. **Answer:** (2)

C begins the paragraph by introducing the topic of discussion, i.e. poverty estimates and the divergence in their estimation. A follows C as it describes an attempt by the NITI Aayog to reduce the divergence (as stated in C) by creating a 'new definition'. B follows A as it questions the need of the 'new methodology'. 'New methodology' in B refers to 'new definition' in A. D concludes the passage by providing an answer to question posed in C, that 'holistic approach' might be better instead of simply following a 'new methodology'. Thus, CABD is the correct sequence.

43. **Answer:** (3)



Clearly, Event II forms the principal and the immediate cause for Event I. The increased access and convenience is a result of Event II.

44. **Answer:** (4)

The text discusses the strength of weak ties in securing jobs. It was believed, wrongly though, that close friends or company head-hunters helped the most. But the study reveals something different. The filler has to reveal the findings. Option 4 tells us that the real source was lesser known contacts. Option 1 brings the extraneous 'communities'; option 2 is about characteristics of people, not them being a resource. Likewise, option 3 is only about the 'ties'.

45. **Answer:** (4)

Option 4 is the best answer. This choice ensures that the introductory phrase "When translating" appears immediately before the pronoun it modifies, "you". Options 1, 2, and 3 are incorrect because "When translating" should appear immediately before the pronoun it modifies, that is, "you", whereas all these choices result in dangling modifiers.

46. **Answer:** (1)

A begins the sequence as it introduces the topic of discussion - 'repeal of Townshend duties'. This 'repeal of Townshend duties' led to 'a relative lull'. C follows A as 'reopened old issues that had been set aside' in C links with 'a relative lull in colonial agitation' in A. Thus, issues that were in 'a relative lull' were 'reopened'. B follows C as 'acceptance of this tea' in B refers to the 'privileged position' given by the Parliament to bail East India Company. This ACB sequence is only evident in option 1.

47. **Answer:** (4)

The paragraph asks a question "why do people fear population ageing?" The answer to it can be given by either option 1 or option 4.

Option 4, in addition to giving the answer, i.e. an aged population becoming a liability also confirms that these fears are not unfounded, i.e. there are reasons for having such fears – frailty, loss of mental acuity (by analogy of individual ageing) and the dependence ratio.

48. **Answer:** (1)

The argument makes a case on the basis of feedback of one agent. The recruitment agent is making a generalisation based on the specific cases observed by him/her, which might or might not be representative of the general trend. This is stated clearly in option (1). Hence, (1) is the valid answer.

49. **Answer:** (2)

The text is about human liking for chilli peppers as part of their diet. The given sentence links the liking to the antimicrobial properties of chilli peppers. The blank begins with 'particularly'. Option 2 provides the answer to 'particularly' - regions where this link needs to be stronger. Option 1 only repeats the information given earlier. Option 3 limits the relation only to fungi. Option 4 doesn't connect with anything in the text.

50. **Answer:** (3)

The text would require a word like 'stop' or 'prevent' since the idea conveyed in the text is that the TRPV1 receptors help us identify how hot a food is and prevent us from eating it. The word closer to 'stop' or 'prevent'



among the options is 'deter'. The TRPV1 receptors don't 'protect', 'separate' or 'refrain' us from eating food that burns.

51. **Answer: (3)**

$$\text{LCM}(3, 4, 6) = 12$$

The highest three-digit multiple of 12 is 996 (12×83), but it is divisible by 83.

999 ($3 \times 3 \times 3 \times 37$) is neither divisible by 2 nor by 83.

984 ($2 \times 2 \times 2 \times 3 \times 41$) is divisible by 3, not by 83.

972 ($2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3$) is divisible by 3, not by 83.

Since we have to take the largest number, 984 is the correct answer.

52. **Answer: (4)**

Since there are 92 roses, 460 tulips, 184 daisies, 207 lilies, and 552 sunflowers, we will get the bouquet size by taking the H.C.F. of 92, 460, 184, 207, and 552.

$$\text{So, H.C.F.} = 23$$

Each bouquet should have 23 flowers.

$$\text{For roses, we need} = 92/23 = 4 \text{ bouquets.}$$

$$\text{For tulips, we need} = 460/23 = 20 \text{ bouquets.}$$

$$\text{For daisies, we need} = 184/23 = 8 \text{ bouquets.}$$

$$\text{For lilies, we need} = 207/23 = 9 \text{ bouquets.}$$

$$\text{For sunflowers, we need} = 552/23 = 24 \text{ bouquets.}$$

$$\text{Minimum number of bouquets required} = 4 + 20 + 8 + 9 + 24 = 65$$

53. **Answer: (5)**

$$16^{203} + 11^{203} \text{ is divisible by } 16 + 11 = 27.$$

So, it is divisible by 9 also.

$$16^{203} \text{ is even and } 11^{203} \text{ is odd.}$$

$$\text{So, } 16^{203} + 11^{203} \text{ is an odd multiple of 9.}$$

So, if it is divided by 18, the remainder will be 9.

54. **Answer: (4)**

A number is divisible by 30 only if it is divisible by 3 and 10, so for divisibility by 10, the unit's place digit should be 0, so $y = 0$.

And for divisibility by 3, the sum of numbers should be divisible by 3.

$$\text{i.e. } 3 + 6 + 5 + x + y = 14 + x \text{ is divisible by 3.}$$

So, the least value of x in this case will be 1.

$$\text{As } 15 = 14 + 1 \text{ is divisible by 3,}$$

$$\text{The least value of } x + y = 1 + 0 = 1$$

55. **Answer: (2)**

$$\text{Ratio of initial investment by Xavier, Yacub and Zeenat} = 3 : 4.5 : 6 = 2 : 3 : 4$$

Let the profit they get be $2a$, $3a$ and $4a$, respectively.

Again, a portion of profit is shared between Yacub and Zeenat in the ratio 1 : 3.



Suppose Yacub and Zeenat get b and $3b$, respectively.

Now, Xavier gets $2a$.

Yacub gets $3a + b$.

Zeenat gets $4a + 3b$.

ATQ, Zeenat's profit share = $1.5 \times$ Yacub's profit share

$$4a + 3b = 1.5 \times (3a + b)$$

$$4a + 3b = 4.5a + 1.5b$$

$$1.5b = 0.5a$$

$$a = 3b \dots (i)$$

$$\text{Total profit} = 18,00,000$$

$$\therefore 2a + 3a + b + 4a + 3b = 18,00,000$$

$$9a + 4b = 18,00,000$$

$$\Rightarrow 27b + 4b = 18,00,000$$

$$\Rightarrow 31b = 18,00,000$$

$$\Rightarrow b = 1800000/31 = 58,064.516$$

$$\text{So, Yacub receives} = 3a + b = 10b = 10 \times 58,064.516 = \text{Rs. } 5,80,645.16.$$

56. **Answer:** (3)

Time taken by Daya to complete the work = 36 days

$$\text{Time taken by Chetan to complete the work alone} = \frac{36}{600} \times 100 = 6 \text{ days}$$

$$\text{Time taken by Birbal to complete the work alone} = \frac{6}{300} \times 100 = 2 \text{ days}$$

$$\text{Time taken by Amit to complete the work alone} = \frac{2}{250} \times 100 = \frac{4}{5} \text{ days} = 0.8 \text{ days}$$

$$\text{Time taken by Jai to complete the work alone} = \frac{0.8}{160} \times 100$$

$$= 1/2 \text{ days} = 0.5 \text{ days}$$

$$\text{Ratio of efficiencies of Jai : Amit : Birbal : Chetan : Daya} = (2) : (5/4) : (1/2) : (1/6) : (1/36) = 72 : 45 : 18 : 6 : 1$$

$$\text{Daya's share} = (1/142) \times \text{total allotted money}$$

$$\text{Rs. } 5,000 = (1/142) \times \text{total allotted money}$$

$$\text{Total allotted money} = \text{Rs. } 5,000 \times 142 = \text{Rs. } 7,10,000$$

57. **Answer:** (3)

Given:

Total number of employees in team A = 35 and total number of employees in team B = 50

The average salary of all the employees in the company = Rs. 60,000

If the average salary of all the employees in team A is 30% more than that of employees in team B



$$\Rightarrow \text{Total number of employees} = 35 + 50 = 85$$

Since the average salary of 85 employees is 60,000

$$\Rightarrow 60,000 = \frac{(\text{Sum of all salaries})}{85}$$

$$\Rightarrow \text{Total salary of all the employees} = 60,000 \times 85 = 51,00,000$$

Let the average salary of team B's employees be b.

$$\Rightarrow \text{Average salary of team A's employees} = b + \left(\frac{30}{100} \times b \right) = \left(\frac{13b}{10} \right)$$

Now,

$$\Rightarrow \text{Total salary} = \text{total number of employees in team A} \times \text{average salary of employees in team A} + \text{total number of employees in team B} \times \text{average salary of employees in team B}$$

$$\Rightarrow 51,00,000 = 45.5b + 50b$$

$$\Rightarrow 95.5b = 51,00,000$$

$$\Rightarrow b = \left(\frac{51,00,000}{95.5} \right) = 53,403.14$$

Hence, approximate average salary of team B's employees = Rs. 53,400

58. **Answer:** (2)

Let Neha's salary be S.

According to the given information, the salary of Priya is 90% more than Neha's salary, which means Priya's salary is $(S + 0.9S) = 1.9S$.

The monthly expenditure of Ritu is one-third of Neha's salary, so Ritu's expenditure is $\left(\frac{1}{3} \right) S$.

The monthly expenditure of Neha is 25% more than that of Ritu, which means Neha's expenditure

$$\text{is } \left(\frac{1}{3} \right) S + 0.25 \left(\frac{1}{3} \right) S = \left(\frac{1}{3} \right) S + \left(\frac{1}{12} \right) S = \left(\frac{5}{12} \right) S.$$

$$\text{Ritu's expenditure} + \text{Neha's expenditure} = \left(\frac{1}{3} \right) S + \left(\frac{5}{12} \right) S = \left(\frac{4}{12} \right) S + \left(\frac{5}{12} \right) S = \left(\frac{9}{12} \right) S$$

Percentage difference = $(\text{Priya's salary} - \text{Total expenditure of Ritu and Neha}) / \text{Priya's salary} \times 100$

$$\text{Percentage difference} = \frac{1.9S - \frac{9}{12}S}{1.9S} \times 100$$

$$\text{Percentage difference} = \frac{1.15S}{1.9S} \times 100$$

Percentage difference $\approx 60.53\%$



Therefore, the total expenditure of Ritu and Neha is approximately 60.53% less than Priya's salary.

59. **Answer:** (4)

Since $y = [(x - 2)^2]^{1/2}$ and $y = (3x + 22)^{1/2}$

$$[(x - 2)^2]^{1/2} = (3x + 22)^{1/2}$$

By squaring both sides, we get

$$(x - 2)^2 = 3x + 22$$

$$x^2 - 4x + 4 = 3x + 22$$

$$x^2 - 7x - 18 = 0$$

$$(x - 9)(x + 2) = 0$$

$$x = 9 \text{ or } -2$$

So, least value of $x = -2$

60. **Answer:** (4)

$$2^{2(x+y)} - 20[2^{x+y}] + 64 \leq 0$$

Assume $2^{x+y} = k$

$$k^2 - 20k + 64 \leq 0$$

$$\Rightarrow (k - 4)(k - 16) \leq 0$$

$$4 \leq k \leq 16$$

$$\Rightarrow 2 \leq x + y \leq 4$$

i.e. $x + y = 2$, only $x = 1$ and $y = 1$ satisfy.

i.e. $x + y = 3$, $x = 1, y = 2$ or $y = 1, x = 2$

i.e. $x + y = 4$, $\Rightarrow (x, y) = (1, 3), (2, 2), (3, 1)$

So, a total of 6 pairs of (x, y) satisfy the given inequality.

61. **Answer:** (4)

The total amount raised by the first five students is: Rs. 37 + Rs. 59 + Rs. 71 + Rs. 83 + Rs. 97 = Rs. 347

To achieve the goal of raising at least Rs. 480, the sixth student must raise at least Rs. 480 - Rs. 347 = Rs. 133.

Since the amount raised by each student must be a prime number, we need to find the smallest prime number that is greater than Rs. 133.

The smallest prime number greater than Rs. 133 is Rs. 137.

Therefore, the minimum amount that the sixth student must raise to ensure that the group achieves their goal of raising at least Rs. 480, while still satisfying the condition that the amount raised must be a prime number, is Rs. 137.

Hence, answer option 4 is correct.

62. **Answer:** (2)

Let P be ' $n \times x$ '.

Therefore, $Q = nx + x$

$$R = nx + 2x$$

$$S = nx + 3x$$



$$T = nx + 4x$$

$$U = nx + 5x$$

$$\text{Sum of the three largest multiples} = S + T + U = nx + nx + nx + 3x + 4x + 5x$$

$$\text{Sum of the three smallest multiples} = P + Q + R = nx + nx + nx + x + 2x$$

$$\text{Difference between the two numbers} = nx + nx + nx + 3x + 4x + 5x - (nx + nx + nx + x + 2x)$$

$$= 3x + 4x + 5x - x - 2x$$

$$= 4x + 5x$$

$$= 9x$$

Hence, answer option 2 is correct.

63. **Answer:** (4)

Let the distance covered in each round be d km.

According to question,

$$\text{Time taken to complete the race at the speed of } 12 \text{ km/h} = \frac{d}{12}$$

$$\text{After that, the runner rested for } 20 \text{ min} = \frac{1}{3} \text{ h.}$$

$$\text{Then he returned to the starting point at the speed of } 18 \text{ km/h} = \frac{d}{18}.$$

$$\text{Time taken for the whole journey from 9 a.m. to 11:05 a.m.} = 125 \text{ min} = \frac{125}{60}$$

$$\left(\frac{d}{12}\right) + \left(\frac{d}{18}\right) + \left(\frac{1}{3}\right) = \left(\frac{125}{60}\right)$$

$$\Rightarrow \left(\frac{d}{12}\right) + \left(\frac{d}{18}\right) = \left(\frac{105}{60}\right)$$

$$\Rightarrow d = \frac{105}{60} \times \frac{216}{30}$$

$$\Rightarrow d = \frac{63}{5} = 12.6 \text{ km}$$

$$\text{Total distance} = 2 \times 12.6 = 25.2 \text{ km}$$

64. **Answer:** (4)

The question is to determine the time taken by Lovely to complete the race.

Since the distance in the given question was 10 km, which remains same,

From statement A:

Let the speed of Lovely be x .

Time taken by Lovely at x speed = t_1

When the speed is increased by 20%, the time taken is t_2 .



From this, we get two equations and by solving, we get the actual time taken by Lovely to complete the race.

From statement B:

Similarly, if the speed decreases by 15%, the time taken will be more than the actual time.

From this, we get two equations and by solving, we get the actual time taken by Lovely to complete the race.

From statement C,

Speed = 8 km/hr

Distance = 10 km

Time taken = $10/8 = 1.25$ hr = 75 minutes

From each of the statements, we can find the time taken by her to finish the race.

So, option 4 is correct.

65. **Answer:** (1)

Speed of Jatin's car = 80 km/hr and speed of Sonia's car = 60 km/hr

According to question,

Both the cars are driving towards each other and cover a distance of 200 km.

So, relative speed = $80 + 60 = 140$ km/hr

Time taken by the cars to meet together,

Time = Distance/Relative speed = $\frac{200}{140} = 1.43$ hours

Further, speed of the bird flying = 90 km/hr

So, distance travelled by the bird = $90 \times 1.43 = 128.7$ km

Therefore, the bird flies approximately 128.7 kilometres before Jatin and Sonia meet.

66. **Answer:** (3)

Given that:

Total number of shares = 4,500

Cost at which Mr. Ganesh bought the shares = Rs. 14

Total cost of all the shares = $4,500 \times 14 = \text{Rs. } 63,000$

Selling price of 3,000 shares = $3,000 \times 18 = \text{Rs. } 54,000$

Selling price of the remaining shares = $1,500 \times 12 = \text{Rs. } 18,000$

Total selling price = $\text{Rs. } 54,000 + \text{Rs. } 18,000 = \text{Rs. } 72,000$

Profit = $\text{Rs. } 72,000 - \text{Rs. } 63,000 = \text{Rs. } 9,000$

67. **Answer:** (2)

Let the cost price be Rs. 100 and the number of units of product be 100 units.

Therefore, marked price = $\frac{120}{100} \times 100 = \text{Rs. } 120$

According to question, he sells 70% of the products, i.e. 70 units, at 10% discount.

Thus, selling price of 70% products = $(70 \times 120) - (0.10 \times 70 \times 120)$
 $= 8,400 - 840$



$$= \text{Rs. } 7,560$$

Later on, he sells the remaining products, i.e. 30 units, at 20% discount.

$$\text{Selling price of the remaining products} = (30 \times 120) - (0.20 \times 30 \times 120)$$

$$= 3,600 - 720 = \text{Rs. } 2,880$$

$$\text{Total selling price} = \text{Rs. } (2,880 + 7,560) = \text{Rs. } 10,440$$

$$\text{Gain} = \text{Selling price} - \text{Cost price} = \text{Rs. } 10,440 - \text{Rs. } (100 \times 100)$$

$$= \text{Rs. } 10,440 - \text{Rs. } 10,000 = \text{Rs. } 440$$

$$\text{Gain\%} = \frac{440}{10,000} \times 100 = 4.4\%$$

68. **Answer:** (3)

Given: Printed price on the watch is Rs. 2,00,000 and successive discounts provided are 35% and 20%.

Also, restoration cost: 25% of the cost price

$$\text{Cost price of the watch after 35\% discount} = \text{Rs. } 2,00,000 - (35\% \text{ of Rs. } 2,00,000) = \text{Rs. } 1,30,000$$

$$\text{Cost price after 20\% discount} = \text{Rs. } 1,30,000 - (20\% \text{ of Rs. } 1,30,000) = \text{Rs. } 1,04,000$$

Restoration cost = 25% of the cost price after the discounts

$$= 25\% \text{ of Rs. } 1,04,000$$

$$= \text{Rs. } 26,000$$

According to question,

$$\text{Total cost price} = \text{Cost price after discounts} + \text{Restoration cost} = \text{Rs. } 1,04,000 + \text{Rs. } 26,000 = \text{Rs. } 1,30,000.$$

$$\text{Intended selling price} = \text{Rs. } 1,30,000 + (50\% \text{ of Rs. } 1,30,000)$$

$$= \text{Rs. } 1,30,000 + (0.50 \times \text{Rs. } 1,30,000)$$

$$= \text{Rs. } 1,95,000$$

$$\text{Final sale price} = \text{Intended selling price} + (\text{Additional offer percentage} \times \text{Intended selling price})$$

$$\text{Final sale price} = \text{Rs. } 1,95,000 + (10\% \text{ of Rs. } 1,95,000)$$

$$= \text{Rs. } 1,95,000 + \text{Rs. } 19,500$$

$$= \text{Rs. } 2,14,500$$

69. **Answer:** (2)

Suppose Dinesh invested at the rate of $R\%$ at State Farm Bank.

$$I = \frac{(P \times R \times T)}{100}$$

$$\text{Or } R = \frac{(16,000 \times 100)}{(64,000 \times 2)}$$

$$R = 12.5$$

$$\text{New rate at Kanara Bank} = 12.5\% + 2.5\% = 15\%$$

$$\text{Total amount invested} = \text{Rs. } (64,000 + 16,000) = \text{Rs. } 80,000$$

Equivalent compound interest of two years at the rate of 15%

$$= \left(15 + 15 + 15 \times \frac{15}{100} \right) \% = 32.25\%$$



$$\begin{aligned}\text{Required compound interest} &= \text{Rs. } 80,000 \times \left(\frac{32.25}{100} \right) \\ &= \text{Rs. } 25,800\end{aligned}$$

Hence, answer option 2 is correct.

70. **Answer:** (1)

Given,

The principal amount and the rate of interest are same for both Brijesh and Varun.

Let P be the principal amount and r be the rate of interest.

Now,

In case of Varun:

$$P \left(1 + \frac{r \times 1}{100} \right) = 4,000 \dots (i)$$

In case of Brijesh:

$$P \left(1 + \frac{r}{100} \right)^2 = 4,160$$

Using equation (i),

$$4,000 \times \left(1 + \frac{r}{100} \right) = 4,160$$

$$r = 4\%$$

Hence, the rate of interest is 4%.

71. **Answer:** (3)

Let 80% of the students who like reading books = A

60% of the students who like watching movies = B

Percentage of the students who enjoy both reading books and watching movies = y%

Since y% represents the portion of students who enjoy both activities, it must be less than or equal to the smaller of the two percentages (reading books or watching movies).

$$\text{So, } y \leq 60 \dots (i)$$

We know that,

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

To minimize $n(A \cap B)$, we have to maximize $n(A \cup B)$ and the maximum value of $n(A \cup B)$ can be 100%.

$$n(A \cap B) \geq 80\% + 60\% - 100\% \geq 40\%$$

So, the required value of y will be $40 \leq y \leq 60$.

72. **Answer:** (3)

$$\text{Given function: } f(x) = \frac{3x^2 - 6x + 12}{x^2 + 2x + 4}$$



$$f'(x) = \frac{(x^2 + 2x + 4)(6x - 6) - (3x^2 - 6x + 12)(2x + 2)}{(x^2 + 2x + 4)^2}$$

$$= \frac{6(x^2 + 2x + 4)(x - 1) - 2(3x^2 - 6x + 12)(x + 1)}{(x^2 + 2x + 4)^2}$$

$$f'(x) = 0$$

$$\frac{3(x^2 + 2x + 4)(x - 1) - (x + 1)(3x^2 - 6x + 12)}{(x^2 + 2x + 4)^2} = 0$$

$$(x^2 + 2x + 4)(x - 1) - (x + 1)(3x^2 - 6x + 12) = 0$$

$$2x^2 - 8 = 0$$

$$x^2 = 4$$

$$x = \pm 2$$

When $x = -2$, the value of the function $\frac{3x^2 - 6x + 12}{x^2 + 2x + 4}$ will be 9.

When $x = 2$, the value of the function $\frac{3x^2 - 6x + 12}{x^2 + 2x + 4}$ will be 1.

So, the maximum value of the function is 9

Therefore, $M = 9$ is the required value.

73. **Answer:** (4)

Since in the group of 13 people, 4 are quadruplets (looking exactly identical), we may have the following choices of selection:

Case 1: When only 4 unique individuals

Case 2: When 1 quadruplet and 3 unique individuals

Case 3: When 2 quadruplets and 2 unique individuals

Case 4: When 3 quadruplets and 1 unique individual

Case 5: When 4 quadruplets and no unique individual

So, effectively, the number of ways of selecting a person: number of ways of case 1 + number of ways of case 2 + number of ways of case 3 + number of ways of case 4 + number of ways of case 5

Since there are $13 - 4 = 9$ unique individuals,

Number of ways of selecting 4 persons = ${}^9C_4 + {}^9C_3 + {}^9C_2 + {}^9C_1 + {}^9C_0$

$$= 126 + 84 + 36 + 9 + 1$$

$$= 256$$

74. **Answer:** (2)



We are given a general formula, for the sum of first n terms of the sequence, with alternating signs.

$$\sum_{i=1}^n (-1)^{i+1} x_i$$

By expanding we get: $= x_1 - x_2 + x_3 - x_4 + \dots + (-1)^{n+1} x_n = n^3 + 3n^2 - 2n$

(Observe that the even terms have a negative sign, while the odd terms have a positive sign.)

$$S_{98} = [98^3 + 3(98)^2 - 2(98)] = 941192 + 28812 - 196 = 969808$$

$$S_{99} = [99^3 + 3(99)^2 - 2(99)] = 970299 + 29403 - 198 = 999504$$

$$S_{97} = [97^3 + 3(97)^2 - 2(97)] = 912673 + 28227 - 194 = 940706$$

$$S_{99} = S_{98} + x_{99}$$

$$x_{99} = S_{99} - S_{98} = 29696$$

$$S_{98} = S_{97} - x_{98}$$

$$x_{98} = S_{97} - S_{98} = -29102$$

$$\text{Hence, } x_{98} + x_{99} = -29102 + 29696 = 594$$

75. **Answer:** (4)

Consider the given equation.

$$\sin^{4\alpha+2} \int_{2\alpha}^{4\alpha} \cos 4x dx = 0$$

$$\sin^{4\alpha+2} \left[\frac{\sin 4x}{4} \right]_{2\alpha}^{4\alpha} = 0$$

$$\sin^{4\alpha+2} [\sin 16\alpha - \sin 8\alpha] = 0$$

$$\sin 4\alpha + \cos 12\alpha \sin 4\alpha = 0$$

$$\sin 4\alpha [1 + \cos 12\alpha] = 0$$

$$\sin 4\alpha = 0$$

$$4\alpha = -\pi$$

$$\alpha = \frac{-\pi}{4}$$

Now,

$$1 + \cos 12\alpha = 0$$

$$\cos 12\alpha = -1$$

$$\cos 12\alpha = \cos (-\pi)$$

$$12\alpha = -\pi$$

$$\alpha = -\frac{\pi}{12}$$



Now,

$$\sin 4^\alpha = 0$$

$$\alpha = 0$$

Hence, option 4 is the correct answer.

76. **Answer:** (2)

$$f(x) = \sin(a \log x) \text{ and } y = f\left(\frac{2x-3}{3x+4}\right)$$

$$\text{Let } \frac{2x-3}{3x+4} = z$$

$$\text{Then } y = f(z)$$

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx}$$

$$\frac{dy}{dz} = f'(z)$$

$$\frac{dz}{dx} = \frac{(3x+4)2 - (2x-3)3}{(3x+4)^2} = \frac{17}{(3x+4)^2}$$

$$f(x) = \sin(a \log x)$$

$$f'(x) = \frac{a \cos(a \log x)}{x}$$

$$f'(z) = \frac{a \cos(a \log z)}{z}$$

$$\frac{dy}{dx} = \frac{a \cos(a \log z)}{z} \times \frac{17}{(3x+4)^2}$$

$$\frac{dy}{dx} = \frac{17a \cos a [\log(2x-3) - \log(3x+4)]}{(2x-3)(3x+4)}$$

$$\text{at } x = 2$$

$$\frac{dy}{dx} = \frac{17a \cos(a \log 10)}{10}$$

77. **Answer:** (3)

$$\log_{x^2+3x+2}(x^2+5x+6) = y$$

$$y = \frac{\log(x^2+5x+6)}{\log(x^2+3x+2)}$$



$$y = \frac{\log(x+3) + \log(x+2)}{\log(x+2) + \log(x+1)}$$

$$\frac{dy}{dx} = \frac{[\log(x+2) + \log(x+1)] \left[\frac{1}{x+3} + \frac{1}{x+2} \right] - [\log(x+3) + \log(x+2)] \left[\frac{1}{x+2} + \frac{1}{x+1} \right]}{[\log(x+2) + \log(x+1)]^2}$$

$$= \frac{[\log(x+2)] \left[\frac{1}{x+3} - \frac{1}{x+1} \right] + \frac{\log(x+1)}{x+3} - \frac{\log(x+3)}{x+1} + \frac{1}{x+2} [\log(x+1) - \log(x+3)]}{[\log(x+2) + \log(x+1)]^2}$$

At $x = 1$,

$$\frac{dy}{dx} = \frac{-\frac{1}{4} \log 3 - \frac{13}{12} \log 2}{[\log 6]^2}$$

$$= -\frac{[13 \log 2 + 3 \log 3]}{12 [\log 6]^2}$$

$$= -\frac{\log 221184}{12 [\log 6]^2}$$

78. **Answer:** (1)

If a 4×4 matrix is filled with entries +2's and -2's such that the sum of each row and column of the matrix is 4, the matrix A can be obtained as:

$$\begin{bmatrix} 2 & 2 & 2 & -2 \\ 2 & 2 & -2 & 2 \\ 2 & -2 & 2 & 2 \\ -2 & 2 & 2 & 2 \end{bmatrix}$$

$$\text{Now, } A = \begin{bmatrix} 2 & 2 & 2 & -2 \\ 2 & 2 & -2 & 2 \\ 2 & -2 & 2 & 2 \\ -2 & 2 & 2 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{bmatrix}$$

By operating $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$ and $R_4 \rightarrow R_4 + R_1$

$$\text{So, } A = 2 \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & -2 & 0 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}$$

Expanding along C_1 , we get

$$|A| = 2 \begin{vmatrix} 0 & -2 & 2 \\ -2 & 0 & 2 \\ 2 & 2 & 0 \end{vmatrix} = 0 \text{ (since the determinant of a skew symmetric matrix is zero)}$$



The value of the determinant of A is $|A| = 0$.

79. **Answer: (2)**

$$\text{Given: } P = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$$

$$\text{Now, } |P| = \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$$

$$= x(yz - x^2) + y(zx - y^2) + z(xy - z^2)$$

$$= 3xyz - x^3 - y^3 - z^3 = -(x^3 + y^3 + z^3 - 3xyz)$$

$$= -[(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)]$$

$$= -\frac{(x + y + z)}{2} [(x - y)^2 + (y - z)^2 + (z - x)^2]$$

= -ve number

Hence, its determinant is always negative.

80. **Answer: (3)**

$$P(\text{he tells a lie}) = \frac{3}{7}$$

$$P(\text{he speaks the truth}) = 1 - \frac{3}{7} = \frac{4}{7}$$

Now, if two dice are thrown, then the observations whose sum is more than 10 are $\{(5, 6), (6, 5), (6, 6)\}$.

Number of observations = 3

The total number of observations is 36.

Now,

$$\text{Probability of getting the sum more than 10} = \frac{3}{36} = \frac{1}{12}$$

$$\text{Probability of not getting the sum more than 10} = 1 - \frac{3}{36} = \frac{33}{36} = \frac{11}{12}$$

The probability that its sum is actually greater than 10 is given by using the Bayes Theorem:

$$\frac{\frac{4}{7} \times \frac{1}{12}}{\frac{4}{7} \times \frac{1}{12} + \frac{3}{7} \times \frac{11}{12}} = \frac{28}{259} = \frac{4}{37}$$



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